



Plantation Journal



**Sustainable Forests & Livelihoods
(SUFAL) Project**

“Mangrove plantation”

Range : Charbata Range

Plantation Area : 500.0 Hectare

Financial Year : 2021-22

Coastal Forest Division, Noakhali

Raising Mangrove of 500.0 Hectare Plantation in Financial Year 2021-2022 under SUFAL Project.

Basic Information :

A.

1 . District	⋮	Noakhali
2 Forest Division	⋮	Coastal Forest Division, Noakhali.
3.Upazila	⋮	Hatiya
4.Range	⋮	Charbata Range
5.Beat	⋮	Bashkhali Beat
6.Union	⋮	Chanondi
7.Name of Char	⋮	Bhasan Char
8.Project	⋮	Sustainable Forests & Livelihoods (SUFAL) Project.
9.Plantation Area	⋮	500.0 Hectare
10.Species	⋮	keora,Gewa.
11.No. of seedlings	⋮	22,22,000 Nos.

B. Characteristic of the Plantation Site		
1	Area of the plantation	500.0 ha. Mangrove Plantation at Bhasan Char the year of 2021-2022.
2	Edaphic Condition	Physiographically the char is part of the young Meghna Estuarine Flood plain. The site (Bhasan Char) is muddy soft and soft banks of small creek as well as canal. these soils permit easy Penetration of Nypa Fruticans (Kere tree roots because of being quite fertile.
3	Climate Condition	The Bhasan Char has a monsoon Climate. Characterized by basically four seasons per year. The average temperature usually ranges between 11 to 29° in winter and 21° to 38°C during summer. The average annual rainfall is 3257 mm and varies from 2500 mm to 4500 mm. Cyclones pass regularly, particularly at the beginning and at the end of the monsoon. The Char is slightly raised flat land which generally inundates twice a day noticeably from March to December. The rate of silt deposition is higher particularly from November to February roughly 3" -4".

4	Brief History	Mangrove plantation site at Bhasan Char is muddy and soft banks of small creeks and canals. The site is the banks of canals that drain water from the central depression of a char. Generally 2 months old seedlings should be taken out of the nursery and planted in the field to minimize root injury. They have been planted at 1.5 m× 1.5m spacing. The site is not subjected to heavy siltation or prone to heavy erosion, while raising keora plantations. In planting Keora strong 1 ft stake has been used against each seedlings planted. Since keora plantations are planted in lines, the funding may be based on “hectare”, planting of 4444 seedlings have been taken as “one seedling hectare.”
5	Selection Criteria of the Planting Site	The plantation site has been selected through the following criteria: • Land suitability: keore plantation site is muddy and some uri grass has grow & it would be very helpful for hard the soil for a established char formation the plantation site is stand allover char . • Grazing Problem: It is known to that there entire are a number of chars in Noakhali and accreting new lands. These chars are threat to the grazing problems. But this is free from such unavoidable phenomenon. That is why this site us selected for the mangrove plantation. • Biodiversity Conservation: The Coastal afforestation program will bring a result in the establishment of plantation rich in biodiversity and higher density of wildlife habitats. Natural regeneration of forest tree species with their associates will be enhanced. In addition, through the project area the condition of coastal forest in Bangladesh will extent. • Sectoral Target: The people residing in the project area are exposed to frequent cyclones and tidal surge which makes them highly vulnerable to recent climate effects. As a result, their socio-economic condition is less secured. Beside this, the land mass developed due to accretion of the sediments discharged through Meghna estuary needs to be stabilized. On the other hand , the objectives of the forestry sector is to increase the total forest cover of the country and to optimize sustainable production of goods and services from the existing limited forest resources. The coastal

		area of the proposed project is such type of land where forestry resources could be developed through establishment of plantations. • National Strategy: The proposed project is also aligned with the national strategy; According to the plan, existing scattered coastal areas will be planted to increase productivity by promoting scientific management principles. In order to prevent the extent of damage by cyclones and tidal surges, the plan envisages creating a Coastal Green Belt and enhancing access to input materials, including seeds and seedlings. In addition, past climate disasters have shown that resilience to cyclones and storm surges is greater where mangrove greenbelts and buffer ecosystems exist. It is evident that mangroves can mitigate or reduce risk of natural disasters such as cyclones and tsunamis. • Global Aspect: According to the IPCC report of 2013, approximately 95% human beings are responsible for the climate change and Bangladesh is the home to the world's largest delta systems. Two third of the country is less than five meters above the sea level. It is vulnerable to short term climate shocks such as cyclone (DFID-2011) while the country is among top ten countries which are to climate change (Maple Craft-2013). As a result, human beings and natural resources are threats to climate change. So, there is no alternative without afforestation and reforestation in the coastal zones. That is why Bhasan Char is selected for raising Keora plantation.
6	Climate Resilient Aspect of the Plantation	Through the afforestation in the project area biodiversity conservation, human security and so on will be ensured in the project area. Therefore, greenbelt of the project area by raising said plantation has been planned to bring back the balance of eco-system into the same char lands and to develop a friendly environment by raising coastal afforestation. It is for the fact that the land is suitable for mangrove and non-mangrove plantations where bio-physical and socio-economic conditions will improve. The Bhasan Char will be brought to work as a barrier during natural disasters by dint of the raised plantation.
C		Other Activities
1	Delineation of the Planting Area	The Bhasan Char is a newly formed char land which has

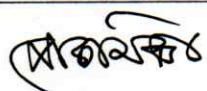
		raised rapidly. The char is adjacent to the East-North side of Keora plantation. The soils of the site	
		Permit easy penetration of keora tree roots because of being fertile. The project site would be used for different types of species diversity plantation. If the coastal forest lands of a target area are properly replenished with greens, the existing wildlife habitats will increase and will be a nice environment for biodiversity conservation. This type of coastal forest lands can fight back the prevailing environmental threats and mitigate the negative impacts of the climate change took place in these coastal zones. Thus, this type of coastal forest lands not only save the life and resources from natural disasters but also play important role in economic development.	
2	GPS Reading of all Corners of the Planting Site	The GPS has been taken from all corners of the plantation site in presence of the RO BO, FG, BM as well as all other staff officer of the coastal Forest Division Noakhali.	1.N 22°23'43.2" E 91°26'43.8" 2.N 22°23'46.9" E 91° 27'00.5" 3.N 22°24'55.9" E 91° 25'50.9" 4.N 22°25'23.4" E 91°26' 33.6" 5.N 22°25'33.4" E 91°25'16.6" 6.N 22°26'05.2" E 91°25'57.5"
3	Post-Planting Survey, Surveyor, Survey Technique & Record	Md. Shafiqul Islam Surveyor, FD-Noakhali The survey has been completed through the technique of GPS reading and it is firstly recorded in the note book after that the record is shown in the Map.	

12. Name & Designation of the Officers & staffs Involved in the Raising Plantation activities :

- A. Mr. Md. Farid Meah , DCF.
- B. Mr. kazi Tariqur Rahman, ACF
- C. Mr. Md. Nazmul Haque Sarder, Forester, Range Officer.
- D. Mr. Moazzem Hossen , Forest Guard, Beat Officer.
- E. Mr. Md. Lokman Hakim, Forest Guard.

- F. Mr. Md. Abdur Rahim, Boat Man.
 G. Mr. Mahabubur Rahman, Boat Man.
 H. Mr. Md. Nur Hossen, Boat Man.
 I. Mr. Md. Mashudur Rahman, Mali.
 J. Mr. Md. Mijan , Mali.
 K. Mr. Md. Shamim , Laskor.

1. Signature of the officers Involved in the Raising Mangrove Plantation activities :

Sl.	Name	Designation	Signature
01	Mr. Moazzem Hosssen Forest Guard	Beat Officer, Bashkhali Beat Charbata Range Coastal Forest Division Noakhali.	 Moazzem Hosssen Beat Officer, Bashkhali Beat Charbata Range Coastal Forest Division, Noakhali.
02	Mr. Md. Nazmul Haque Sarder Forester	Range Officer Charbata Range Coastal Forest Division Noakhali.	 Md. Nazmul Haque Sarder Range Officer Charbata Range Coastal Forest Division, Noakhali.
03	Mr. Kazi Tariqur Rahman, ACF	Assistant Conservator of Forest Coastal Forest Division Noakhali.	 Kazi Tariqur Rahman Assistant Conservator of Forest Coastal Forest Division Noakhali.
04	Mr. Md. Farid Meah, DCF.	Divisional Forest Officer Coastal Forest Division Noakhali.	 Md, Farid Meah Divisional Forest Officer Coastal Forest Division Noakhali.

EXPENDITURE SHEET

Range : Charbata Range, Coastal Forest Division, Noakhali.
Project : Sustainable Forests & Livelihoods (SUFAL) Project

Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
10/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	25,000/-	10/643
11/Chaba	December/2021	Do	25,000/-	50,000/-	11/643
12/Chaba	December/2021	Do	25,000/-	75,000/-	12/643
13/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 550 Nos @ 500/-	2,75,000/-	3,50,000/-	13/643
14/Chaba	December/2021	Do Daily Labours 550 Nos @ 500/-	2,75,000/-	6,25,000/-	14/643
15/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	6,50,000/-	15/643
16/Chaba	December/2021	Do	25,000/-	6,75,000/-	16/643
17/Chaba	December/2021	Do	25,000/-	7,00,000/-	17/643
18/Chaba	December/2021	Do	25,000/-	7,25,000/-	18/643
19/Chaba	December/2021	Do	25,000/-	7,50,000/-	19/643
20/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 550 Nos @ 500/-	2,75,000/-	10,25,000/-	20/643
21/Chaba	December/2021	Do Daily Labours 550 Nos @ 500/-	2,75,000/-	13,00,000/-	21/643


Md. Nazmul Haque Sarder
 Range Officer
 Charbata Range
 Coastal Forest Division, Noakhali.

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Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
22/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	13,25,000/-	22/643
23/Chaba	December/2021	Do	25,000/-	13,50,000/-	23/643
24/Chaba	December/2021	Do	25,000/-	13,75,000/-	24/643
25/Chaba	December/2021	Do	25,000/-	14,00,000/-	25/643
26/Chaba	December/2021	Do	25,000/-	14,25,000/-	26/643
27/Chaba	December/2021	Do	25,000/-	14,50,000/-	27/643
28/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 550 Nos @ 500/-	2,75,000/-	17,25,000/-	28/643
29/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	17,50,000/-	29/643
30/Chaba	December/2021	Do	25,000/-	17,75,000/-	30/643
31/Chaba	December/2021	Do	25,000/-	18,00,000/-	31/643
32/Chaba	December/2021	Do	25,000/-	18,25,000/-	32/643
33/Chaba	December/2021	Do	25,000/-	18,50,000/-	33/643
34/Chaba	December/2021	Do	25,000/-	18,75,000/-	34/643
35/Chaba	December/2021	Do	25,000/-	19,00,000/-	35/643
36/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 550 Nos @ 500/-	2,75,000/-	21,75,000/-	36/643



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Project : Sustainable Forests & Livelihoods (SUFAL) Project

Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
37/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	22,00,000/-	37/643
38/Chaba	December/2021	Do	25,000/-	22,25,000/-	38/618
39/Chaba	December/2021	Do	25,000/-	22,50,000/-	39/618
40/Chaba	December/2021	Do	25,000/-	22,75,000/-	40/618
41/Chaba	December/2021	Do	25,000/-	23,00,000/-	41/618
42/Chaba	December/2021	Do	25,000/-	23,25,000/-	42/618
43/Chaba	December/2021	Do	25,000/-	23,50,000/-	43/618
44/Chaba	December/2021	Daily labours were engaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 520 Nos @ 500/-	2,60,000/-	26,10,000/-	44/618
45/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	26,35,000/-	45/643
46/Chaba	December/2021	Do	25,000/-	26,60,000/-	46/643
47/Chaba	December/2021	Do	25,000/-	26,85,000/-	47/643
48/Chaba	December/2021	Do	25,000/-	27,10,000/-	48/643
49/Chaba	December/2021	Do	25,000/-	27,35,000/-	49/643
50/Chaba	December/2021	Do	25,000/-	27,60,000/-	50/643
51/Chaba	December/2021	Daily labours were engaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 520 Nos @ 500/-	2,60,000/-	30,20,000/-	51/643


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Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
52/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	30,45,000/-	52/643
53/Chaba	December/2021	Do	25,000/-	30,70,000/-	53/643
54/Chaba	December/2021	Do	25,000/-	30,95,000/-	54/643
55/Chaba	December/2021	Do	25,000/-	31,20,000/-	55/643
56/Chaba	December/2021	Do	25,000/-	31,45,000/-	56/643
57/Chaba	December/2021	Do	25,000/-	31,70,000/-	57/643
58/Chaba	December/2021	Do	25,000/-	31,95,000/-	58/643
59/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 520 Nos @ 500/-	2,60,000/-	34,55,000/-	59/643
60/Chaba	December/2021	Boat hire for transportation of seedlings to the plantation site form nursery site	25,000/-	34,80,000/-	60/643
61/Chaba	December/2021	Do	25,000/-	35,05,000/-	61/643
62/Chaba	December/2021	Do	25,000/-	35,30,000/-	62/643
63/Chaba	December/2021	Daily labours were angaged for Survey the plantation site and prepare plantation site map and plantation journal, uprooting of the seedlings at the nursery side, strip of the lower leaps leaving 3 to 4 top leaves, prune the excessive lateral roots, place them systematically on wet hogla mat for loading on the boat for transportation, of seedlings to plantation side mostly by boat including loading and unloading, cleaning of Dhanshi inf lines and planting of the bare rooted seedlings carefully at 1.5 m× 1.5m spacing for rising of mangrove Plantation under SUFAL Project. Daily Labours 140 Nos @ 500/-	70,000/-	36,00,000/-	63/643
		Cost of Delivery of goods by the Contractor =	1,75,000/-		
		Total cost of 500.0 Hectare =	37,75,000/-		


 Md, Nazmul Haque Sarder
 Range Officer
 Charbata Range
 Coastal Forest Division, Noakhali.

" Sustainable Forests & Livelihoods (SUFAL) Project "

Post Planting Map

Coastal Forest Division, Noakhali.

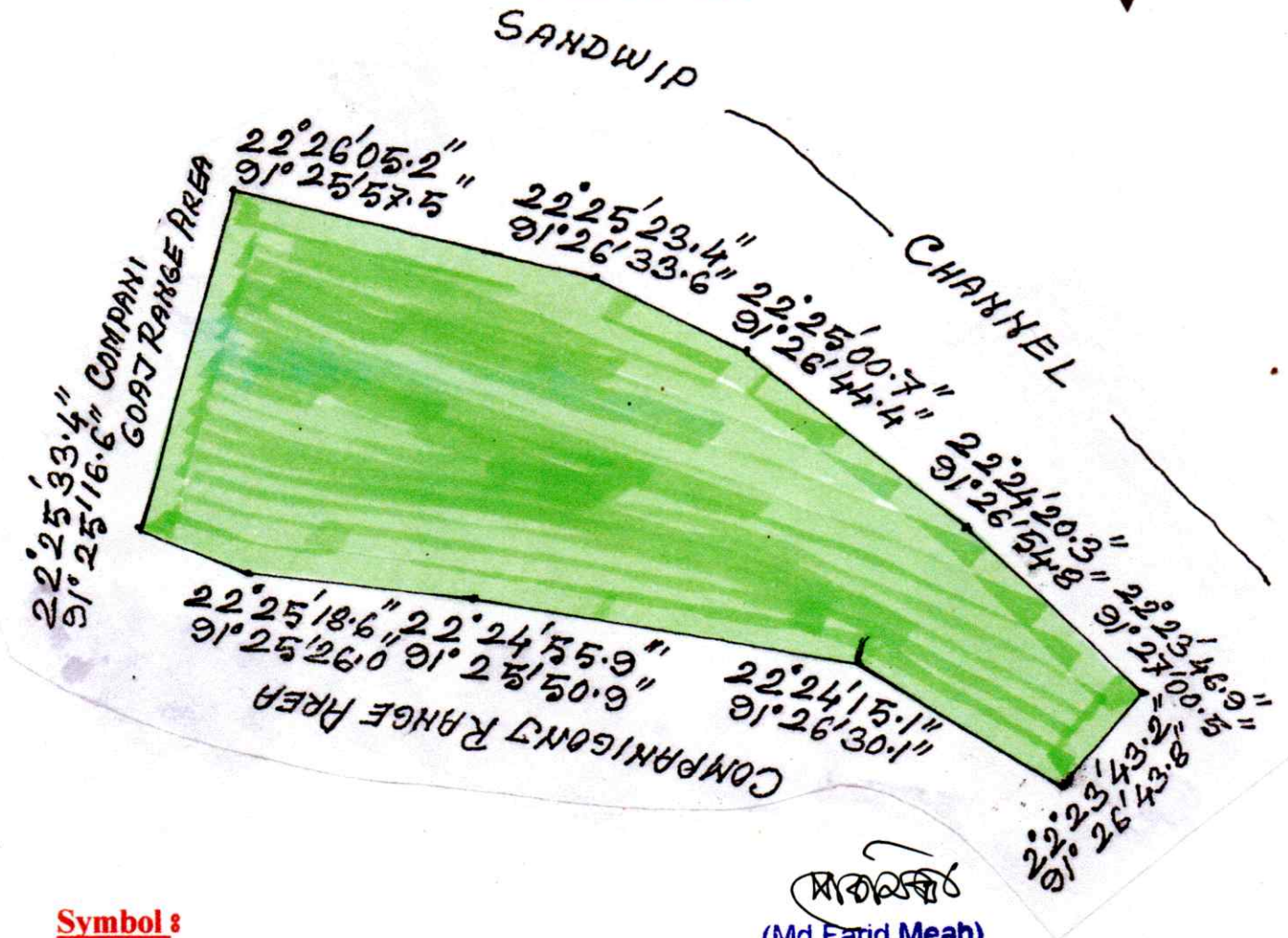
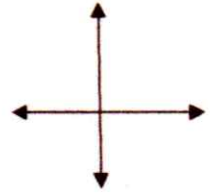
Beat : Bashkhali, Range: Charbata

Name of the Char : Bhasan Char

Plantation Area : 500.0 Hectare

Financial Year : 2021-2022

Scale : 1: 31680



Symbol :

- ☆ Plantation Area 2021-2022 : ☐
- ☆ River : ☐

(Md Farid Meah)

Deputy Conservator of forests
Divisional Forest Officer
Coastal Forest Division, Noakhali.

26/11

Counter Signed

Submitted by

Prepared by


Kazi Tariqur Rahman
Assistant Conservator of
Forest Coastal Forest Division
Noakhali.

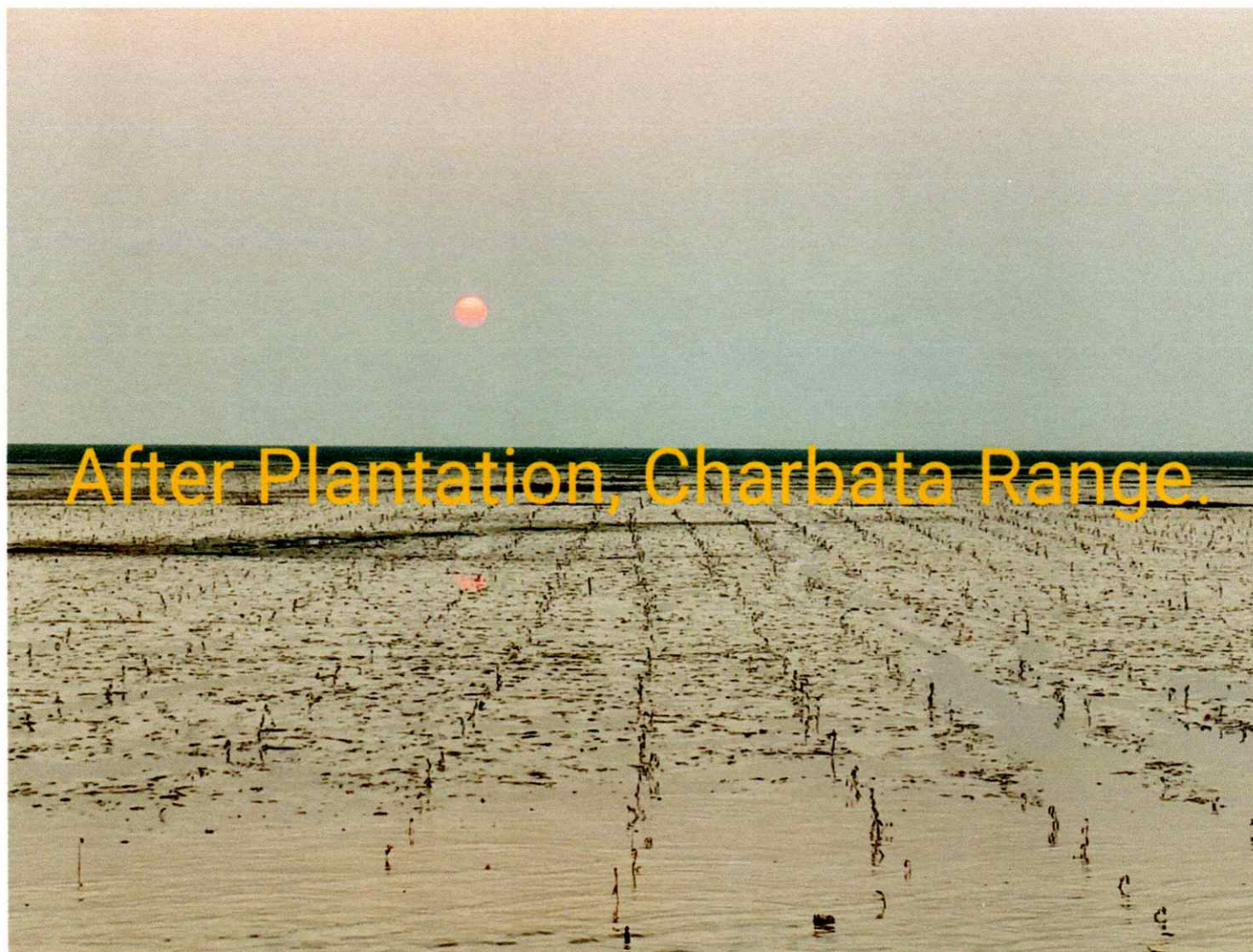

Md. Nazmul Haque Sarder
Range Officer
Charbata Range
Coastal Forest Division,
Noakhali.


Moazzem Hossen
Beat Officer, Bashkhali Beat
Charbata Range
Coastal Forest Division,
Noakhali.


Md. Shafiqul Islam
Surveyor
Coastal Forest Division
Noakhali.



**Md. Nazmul Haque Sarder
Range Officer
Charbata Range
Coastal Forest Division, Noakhala**



Md. Nazmul Haque Sarder
Range Officer
Charbata Range
Coastal Forest Division, N. B. P.

After plantation, Charbata Range.



Md. Nazmul Haque Sarker
Range Officer
Charbata Range
Coastal Forest Division, Noakhata

SUFAL Project. After plantation, Charbata Range.



Md. Nazmul Haque Sarder
Range Officer
Charbata Range
Coastal Forest Division, Noakhata